CIS 9650 Project

Programming for Analytics

US 2023 Civil flights and delays

Group 8

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US FLIGHTS AND CANCELLED FLIGHTS DATA IN THE YEAR 2023

RAW DATA - PREVIEW

```
flights_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6743404 entries, 0 to 6743403
Data columns (total 24 columns):
 #   Column                Dtype
---  ------                -----
 0   FlightDate            object
 1   Day_Of_Week           int64
 2   Airline               object
 3   Tail_Number           object
 4   Dep_Airport           object
 5   Dep_CityName          object
 6   DepTime_label         object
 7   Dep_Delay             int64
 8   Dep_Delay_Tag         int64
 9   Dep_Delay_Type        object
 10  Arr_Airport           object
 11  Arr_CityName          object
 12  Arr_Delay             int64
 13  Arr_Delay_Type        object
 14  Flight_Duration       int64
 15  Distance_type         object
 16  Delay_Carrier         int64
 17  Delay_Weather         int64
 18  Delay_NAS             int64
 19  Delay_Security        int64
 20  Delay_LastAircraft    int64
 21  Manufacturer          object
 22  Model                 object
 23  Aicraft_age           int64
dtypes: int64(11), object(13)
```

```
disrupted_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 104488 entries, 0 to 104487
Data columns (total 23 columns):
 #   Column                Non-Null Count   Dtype
---  ------                --------------   -----
 0   FlightDate            104488 non-null  object
 1   Day_Of_Week           104488 non-null  int64
 2   Airline               104488 non-null  object
 3   Tail_Number           104488 non-null  object
 4   Cancelled             104488 non-null  float64
 5   Diverted              104488 non-null  float64
 6   Dep_Airport           104488 non-null  object
 7   Dep_CityName          104488 non-null  object
 8   DepTime_label         104488 non-null  object
 9   Dep_Delay             104488 non-null  float64
 10  Dep_Delay_Tag         104488 non-null  int64
 11  Dep_Delay_Type        104488 non-null  object
 12  Arr_Airport           104488 non-null  object
 13  Arr_CityName          104488 non-null  object
 14  Arr_Delay             104488 non-null  float64
 15  Arr_Delay_Type        104488 non-null  object
 16  Flight_Duration       104488 non-null  float64
 17  Distance_type         104488 non-null  object
 18  Delay_Carrier         104488 non-null  float64
 19  Delay_Weather         104488 non-null  float64
 20  Delay_NAS             104488 non-null  float64
 21  Delay_Security        104488 non-null  float64
 22  Delay_LastAircraft    104488 non-null  float64
dtypes: float64(10), int64(2), object(11)
```

Glimpse into our datasets

- Number of duplicates: 31 & 394
- Merged the two dataframes on 'FlightDate', 'Day_Of_Week', 'Airline', 'Tail_Number', 'Dep_Airport'

```
Number of columns in the flights data: 24
Number of rows in the flights data: 6743404
```

```
Number of columns in the disrupted data: 23
Number of rows in the disrupted data: 104488
```

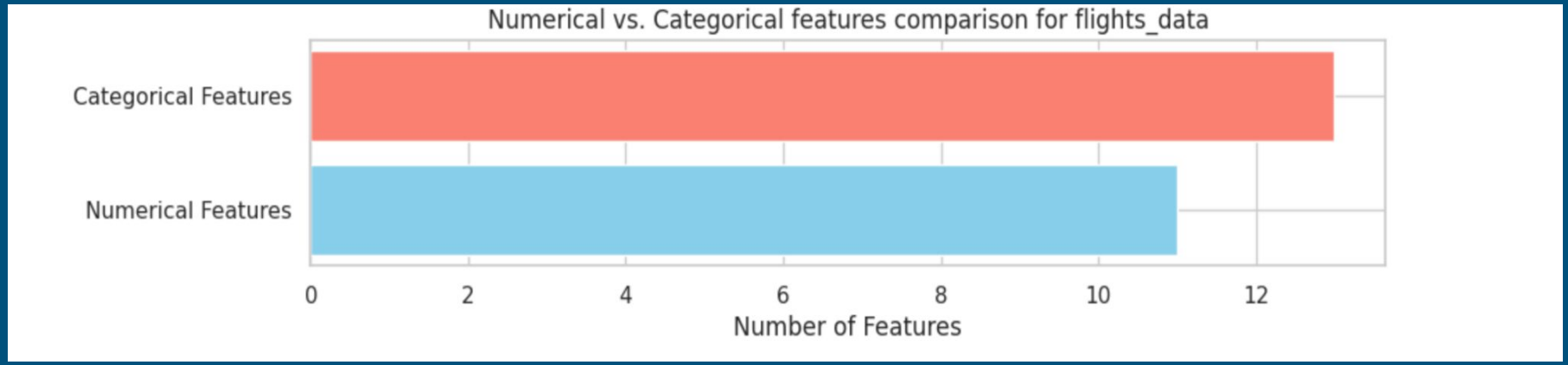
```
Shape of filtered_flights: (1333667, 12)
Original Shape of flights_data: (6743404, 24)
```

```
Shape of filtered_disrupted: (7745, 12)
Original Shape of disrupted_data: (104488, 23)
```

Cleaning Up Our Data

1. STARTED WITH 4 DATASETS AND ENDED UP NARROWING IT DOWN TO 2 DATASETS
2. FOUND OUT DUPLICATES IN THE DATA AND REMOVED THEM
3. DEALT WITH THE NULL VALUES
4. REMOVED THE UNNECESSARY COLUMNS
5. CHANGED COLUMN TYPE

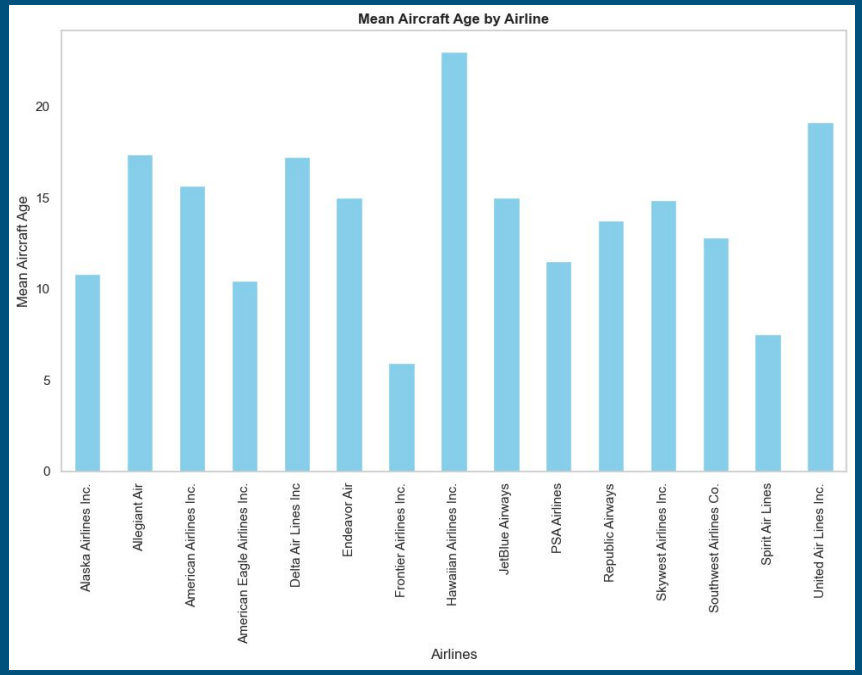
Preliminary EDA On Unfiltered Data



- Categorical data is often easier to interpret as it represents discrete categories or labels and aids in understanding the distribution of observations across different categories, facilitating exploratory data analysis.
- Numerical data allows for quantitative analysis, enabling mathematical operations such as mean, median, standard deviation, etc.

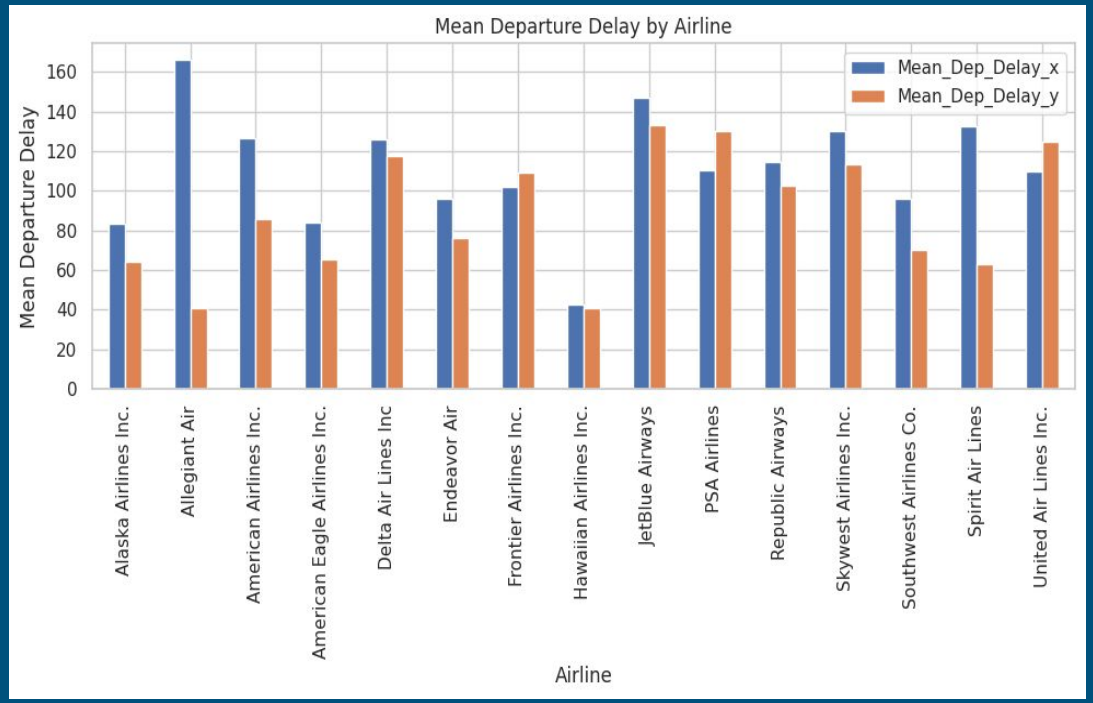
Statistical Analysis

Aircraft age (years)-
Count - 1074.000000
Mean - 14.412477
Std - 7.858741
Min - 1.000000
25% - 8.000000
50% - 14.000000
75% - 21.000000
Max - 48.000000



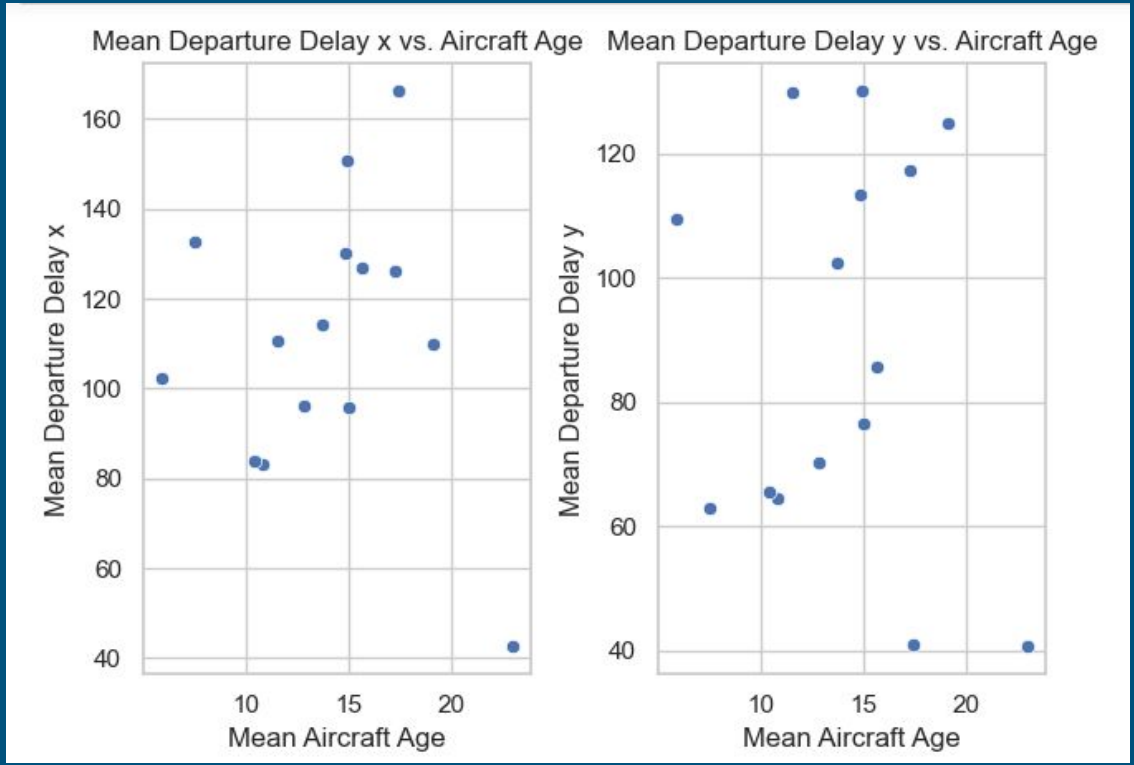
Understanding the Mean Departure Delay

- Studying ‘dep_delay’ values from both the datasets for any patterns.
- Both sets show similar trends in delay patterns across airlines but with varying magnitudes.
- JetBlue Airways and Spirit Airlines Co. also show relatively higher delays across both datasets.
- Delta Air Lines Inc. and Hawaiian Airlines Inc. tend to have lower delays, indicating potentially better on-time performance.



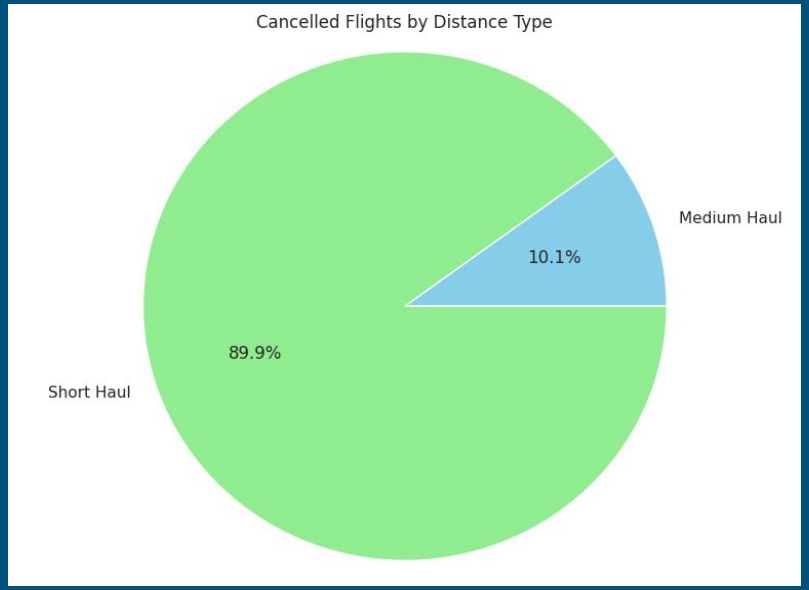
Correlation between Mean_Dep_Delay and Aircraft_age

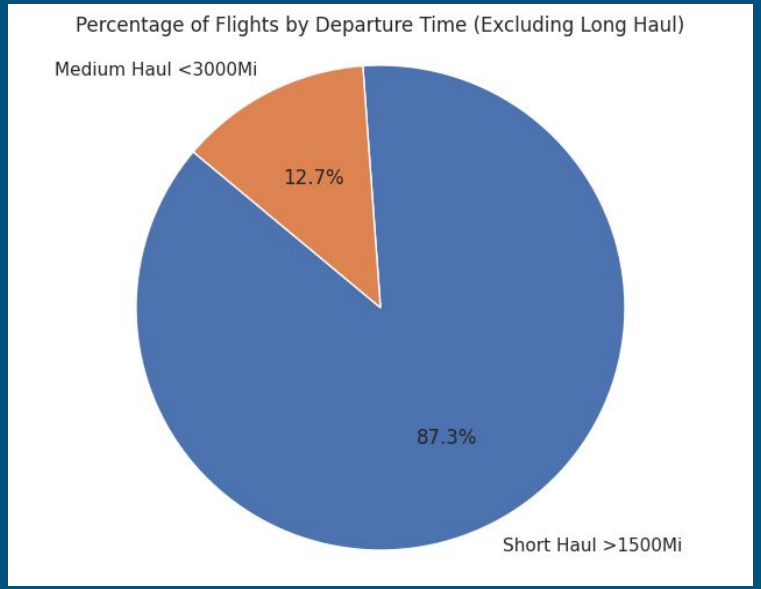
- Is age not a major determinant?
- Spread of data points that does not clearly follow a linear trend
- Indicating that other factors beyond aircraft age might be influencing departure delays.
- Data collection periods, operational conditions, or delay reporting/calculation methods between the two datasets.



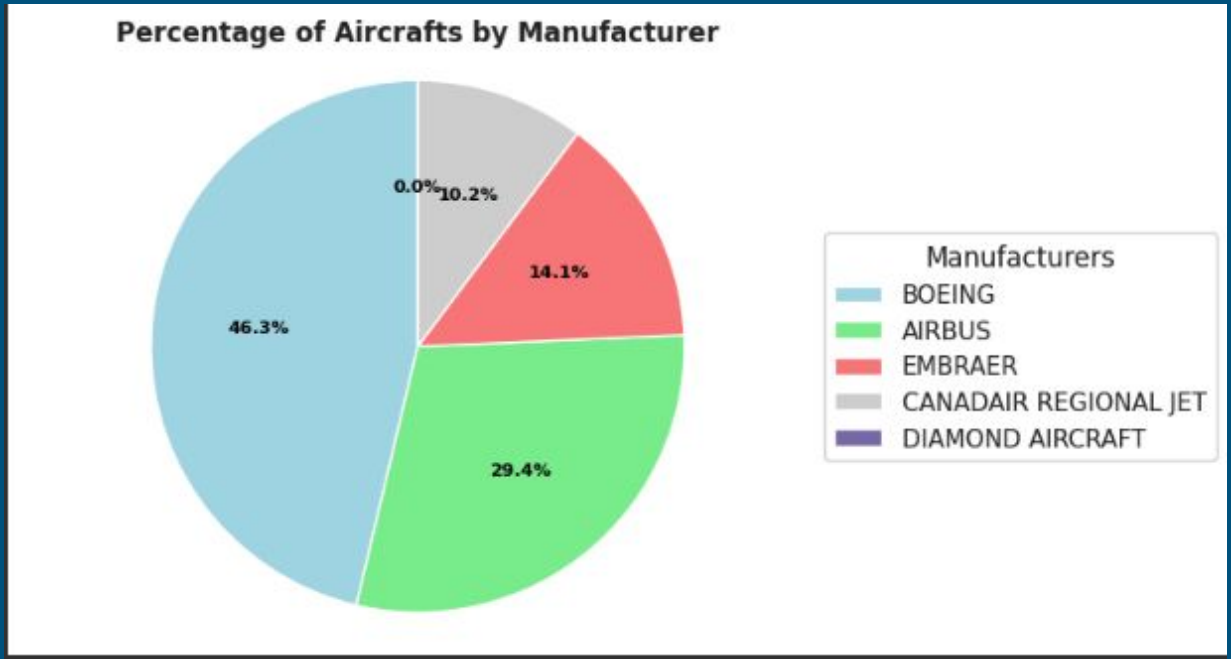
Flights analysis as per run_time categories

We decided to further narrow down the approach by zooming in the cases where departure delays are either 'Medium' and short because long_haul only accounted for less than 1% percent





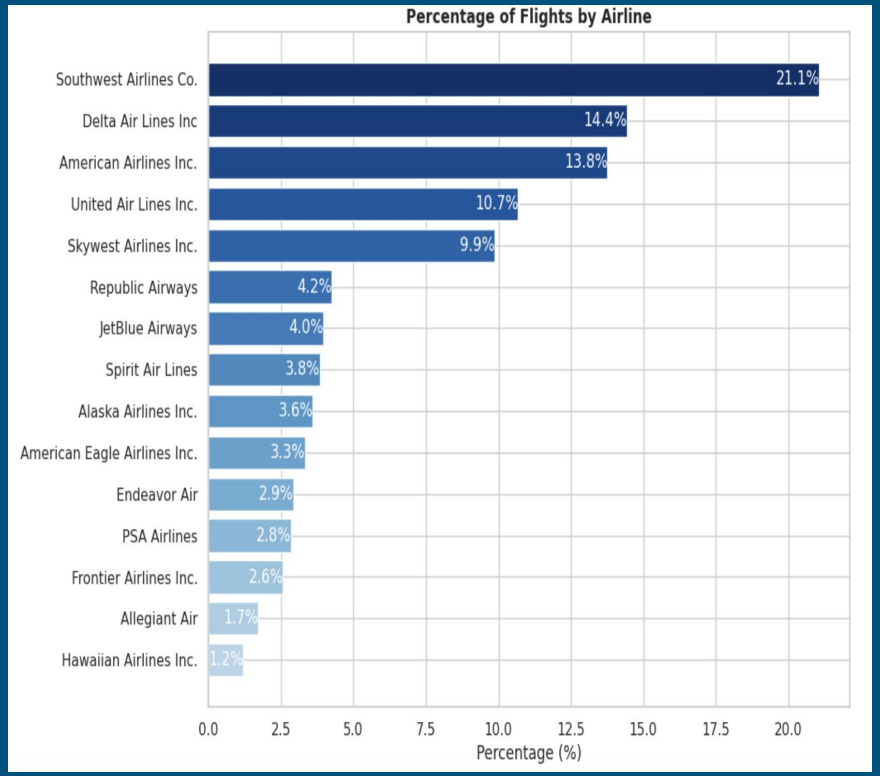
Who is the Largest Airlines Manufacturer for the US in 2023?

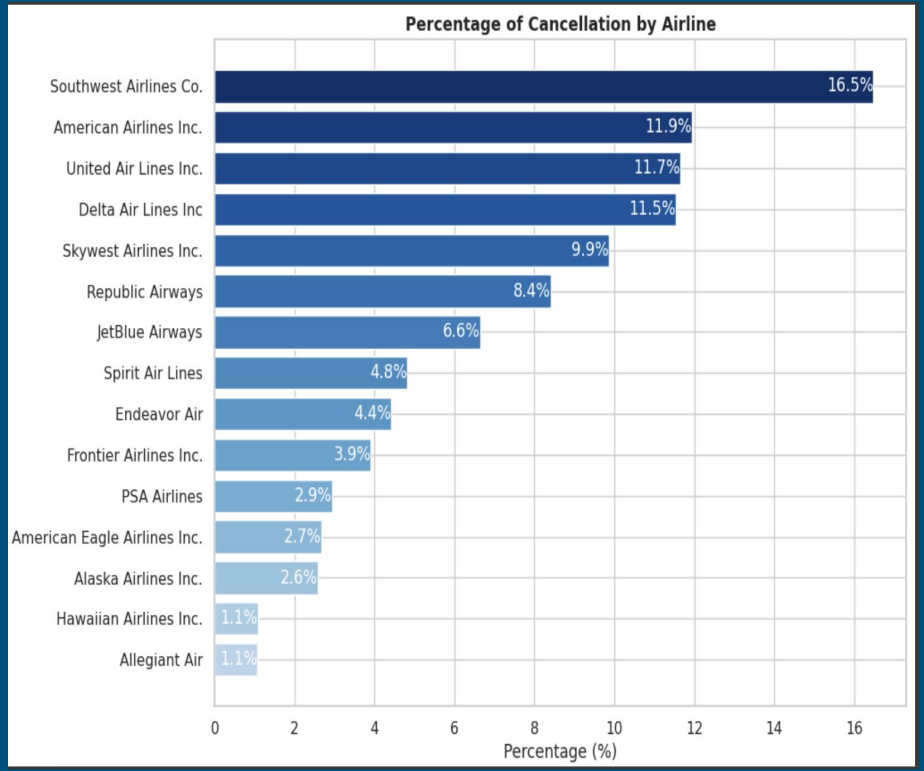


Boeing is the dominant player in the industry with almost half a share of the industry.

Airbus with the second largest market share is acting as a strong competitor for Boeing.

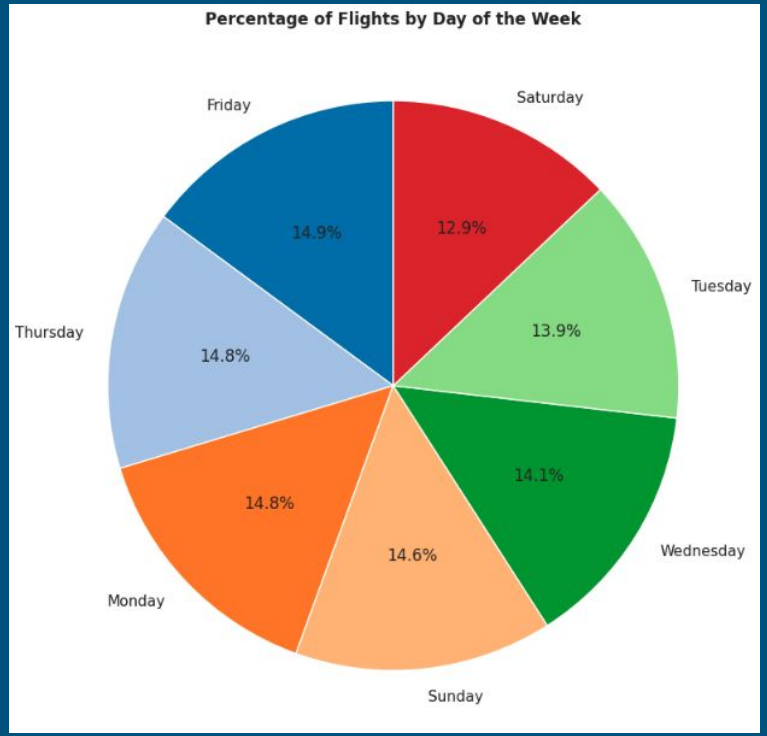
Comparison Between Operating and Canceled Flights by Airline in 2023

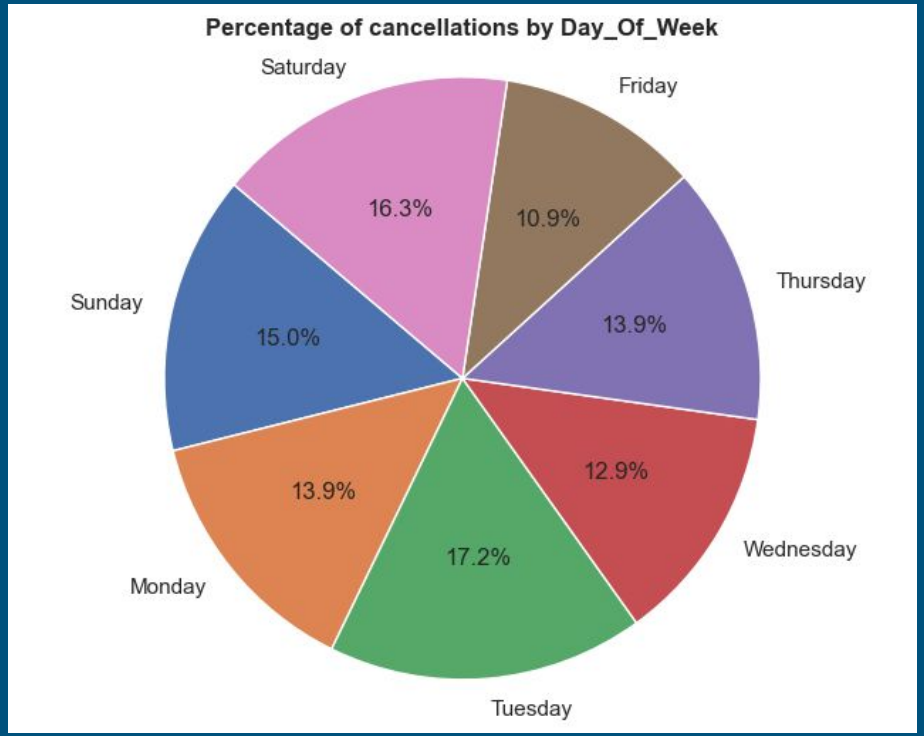




Southwest Airlines Co. is the largest airlines in the United States in terms of the number of flights operated as well as canceled. Following its lead Delta has the second highest operated flights whereas American Airlines has the second highest percent of canceled flights.

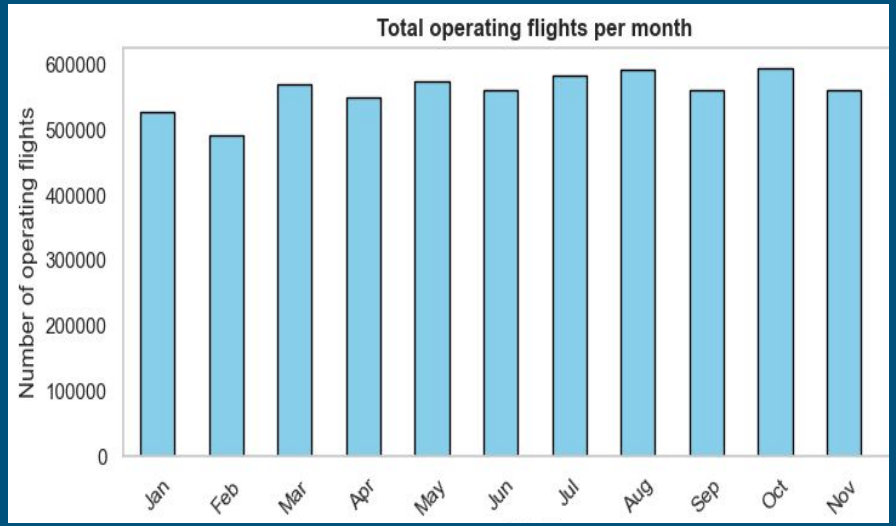
Comparison Between Operating and Canceled Flights by Day of the Week

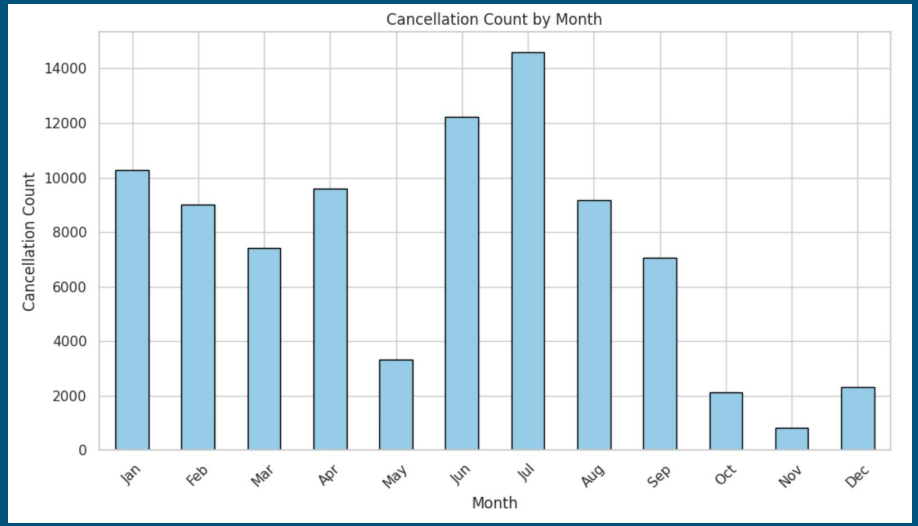




From the above charts, we can infer that there is not much difference between percent of operating flights everyday.
We also see that Tuesdays have the highest percent of canceled flights whereas Fridays has the least.

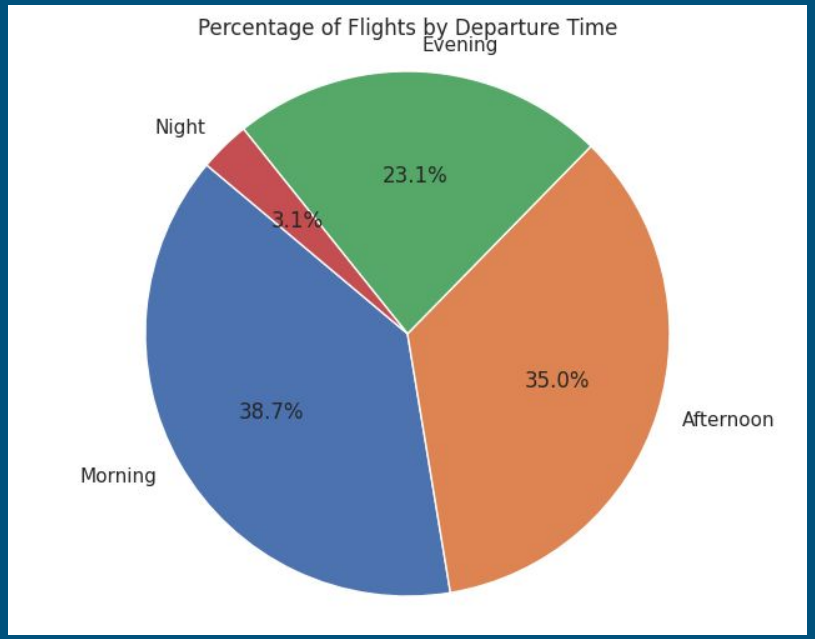
Comparison Between Operating and Canceled Flights by Month

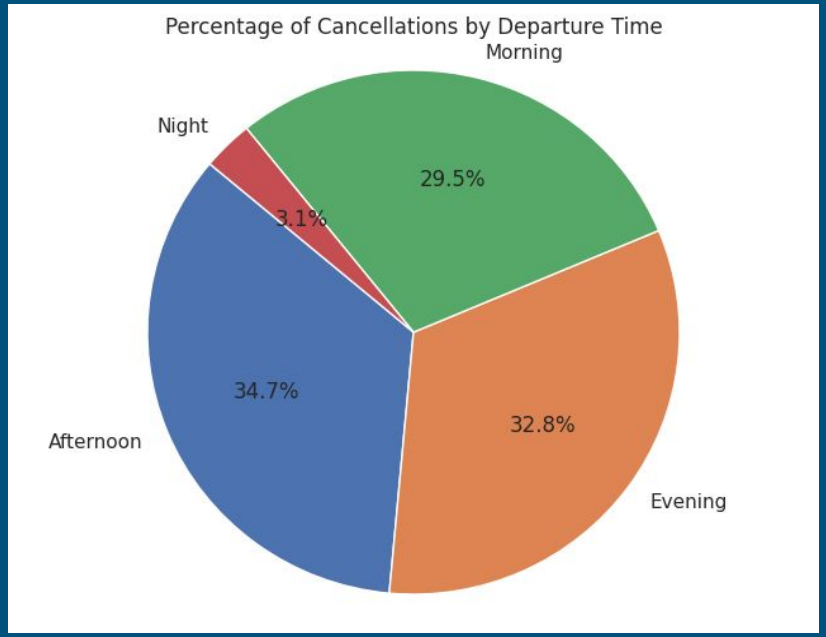




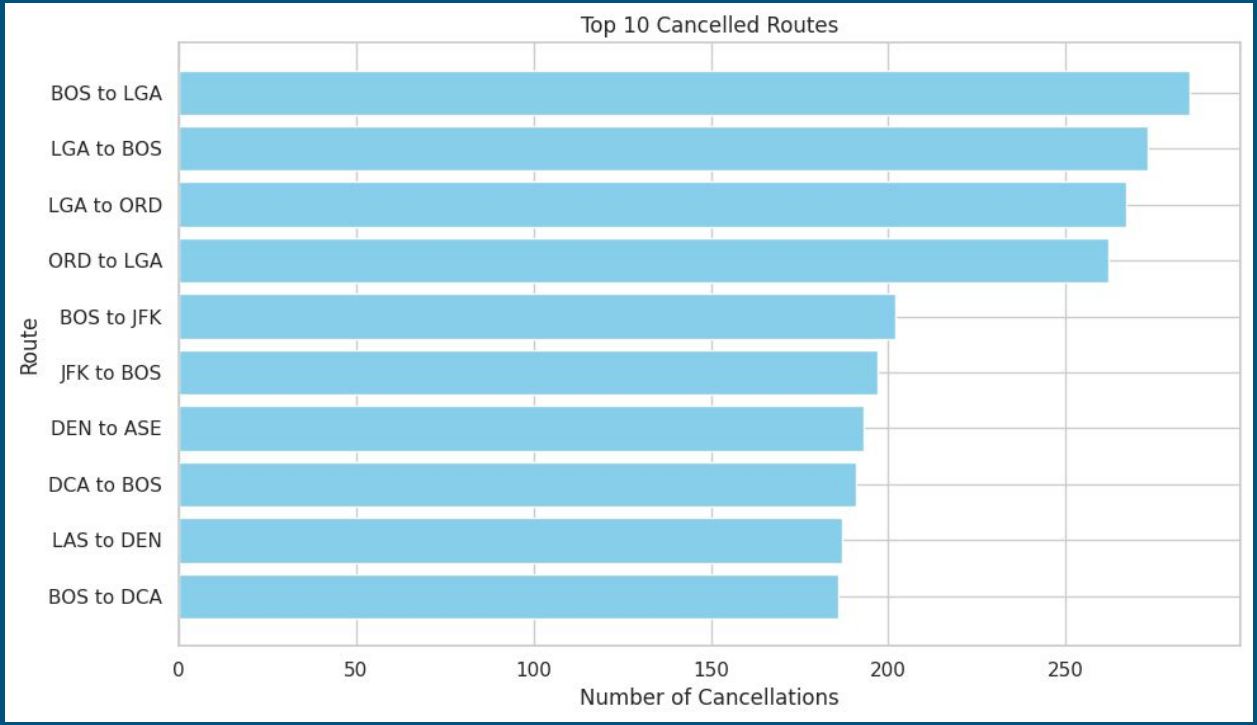
From the above charts, we can infer that there is not much difference between percent of operating flights every month. October and August have most operating flights with February being the lowest.
We also see that July have the highest percent of canceled flights whereas November has the least.

Comparison Between Operating and Canceled Flights by Departure Time





Top 10 Cancelled Routes



From the above chart, we can infer that the Boston Logan Airport to LaGuardia has the highest amount of cancellations.

Beneficiaries of our Analysis

- Analysts, researchers, and policymakers
- Airlines to understand and address cancellation pattern

Challenges Faced:

- Volume and Complexity
- Missing Data
- Data Integration

Key Findings

01	Largest Manufacturer in US	Boeing- 46% market share.
02	Largest Airline in 2023	Southwest Airlines Co.
03	Month, day and time with highest cancellations	-June and July -Saturday and Tuesday -Afternoons
04	Route with most cancellations	The most canceled route was Boston Logan Airport To LaGuardia.
05	Type of most canceled flights	89.7% - small-haul.

Thank You